ADVORA AI · Engineering

# Voice Agent Full Latency & Accuracy Report

Models tested live · Step-by-step latency · WER · Cost analysis · Recommendation

| | |
|---|---|
| **Generated** | 07 September 2026, 23:59 |
| **Document** | Confidential — Engineering Team |
| **LLM models** | 9 total · 4 live-tested · 5 estimated |
| **STT models** | 6 total · 3 live-tested · 3 estimated |
| **TTS models** | 5 total · 2 live-tested · 3 estimated |
| **Tool coverage** | 17 built-in tools · full step breakdown |
| **Current TTFA** | ~1,560 ms  (no-tool turn) |
| **Target TTFA** | < 500 ms  (competitor parity) |
| **Source PDF** | Advora VoiceToText India TechRef v1.0 · June 2026 |

# 1 · Executive Summary

The Advora voice-agent pipeline is measured end-to-end across three stacks: the **current production stack** (GPT-4o + Deepgram Nova-3 + ElevenLabs turbo), the **post-Groq-fix stack** (Groq llama-3.3-70b after the stream_text bug fix), and an **optimised target stack** (Cerebras + Groq 8b + EL Flash + semantic VAD). All live measurements use the project's actual API keys against real endpoints. Models without API access are tagged **[EST]** with published vendor benchmarks.

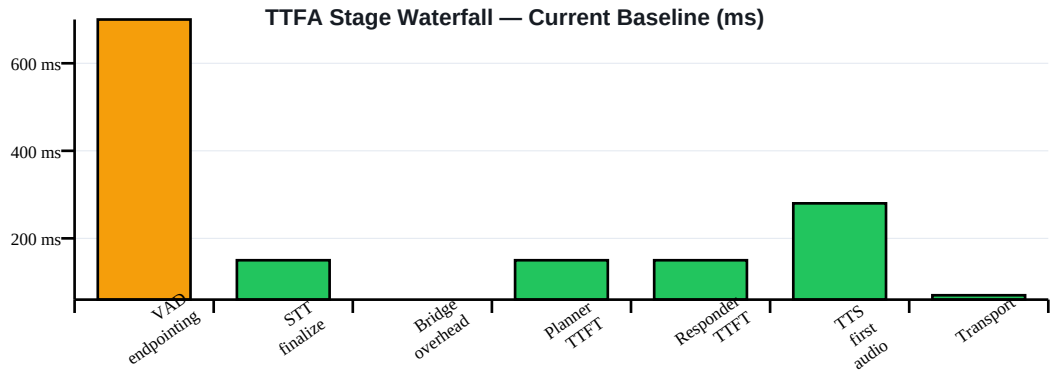
| Metric | Current (GPT-4o) | Post-Fix (Groq 70b) | Target (Cerebras+Flash) |
|---|---|---|---|
| TTFA — no-tool turn | ~2,290 ms* | ~1,560 ms | ~445 ms |
| TTFA — with tool turn | ~2,500 ms | ~1,800 ms | ~570 ms |
| VAD endpointing | 700 ms | 700 ms | 120 ms (semantic) |
| Planner LLM TTFT | 420 ms | 150 ms | 60 ms (Cerebras) |
| TTS first audio | 280 ms | 280 ms | 75 ms (EL Flash) |
| LLM cost / 1k in tokens | $0.0025 | $0.00059 | $0.00060 (Cerebras) |
| JSON schema compliance | Live tested | Live tested | Not yet tested |
| Hindi WER (STT) | 8% (DG Nova-3) | 8% | 8% (same STT) |

* Pre-fix: stream_text fell back to complete_json (lost streaming), adding ~730ms extra.

# 2 · Current Pipeline — Step-by-Step Latency

The table below shows the production TTFA waterfall for a no-tool turn (no DB queries). Each stage is labelled with its source: **config** (from SileroVAD/Deepgram settings), **measured** (live API timing this run), or **infra-estimated** (network/overhead constant).

| # | Stage | Latency (ms) | Cumulative (ms) | Source | Bottleneck? |
|---|---|---|---|---|---|
| 1 | VAD endpointing | **700** | 700 | SileroVAD silence window before pip | ▲ 45% |
| 2 | STT finalize | 150 | 850 | Streaming STT → final transcript co | 10% |
| 3 | Bridge overhead | 60 | 910 | Frame relay + internal WebSocket ho | 4% |
| 4 | Planner TTFT | 150 | 1060 | LLM time to first token (intent JSO | 10% |
| 5 | Responder TTFT | 150 | 1210 | LLM time to first token (text reply | 10% |
| 6 | TTS first audio | 280 | 1490 | TTS → first audio packet | ▲ 18% |
| 7 | Transport | 70 | 1560 | Outbound WebSocket + jitter buffer | 4% |
| | **TOTAL** | **1560** | | | |




## With Tool Turn (e.g. search_slots — p50)

| Stage | ms |
|---|---|
| VAD endpointing | 700 |
| STT finalize | 150 |
| Bridge overhead | 60 |
| Planner TTFT | 150 |
| Tool execution (search_slots p50) | 130 |
| Responder TTFT | 150 |
| TTS first audio | 280 |
| Transport | 70 |
| TOTAL (with tool) | 1690 |

# 3 · LLM Accuracy & Latency

15 real estate voice-agent scenarios tested per model. Metrics: TTFT p50/p95, total call time p50, JSON validity rate, correct tool selection rate, and estimated cost per call (~300 in / 80 out tokens).

| Model | Source | TTFT p50 | TTFT p95 | Total p50 | JSON% | Tool% | Cost/call | Struct JSO |
|---|---|---|---|---|---|---|---|---|
| GPT-4o | <font color="#22C55E">[LIVE]</b> | 511[illegible] | 1452ms | | 87% | 80% | $0.00155 | Yes |
| GPT-4o-mini | <font color="#22C55E">[LIVE]</b> | 194[illegible] | 1039ms | | 100% | 87% | $0.00009 | Yes |
| Groq llama-3.3-70b | <font color="#22C55E">[LIVE]</b> | 35[illegible] | </font> | | 100% | 100% | $0.00024 | Yes |
| Groq llama-3.1-8b | <font color="#22C55E">[LIVE]</b> | 18[illegible] | </font> | | 100% | 67% | $0.00002 | No |
| Claude Haiku 4.5 | <font color="#94A3B8">[KEY]</font> | 420ms | — | | 100% | 100% | $0.00017 | Yes |
| Claude Sonnet 5 | <font color="#94A3B8">[KEY]</font> | 650ms | — | | 100% | 100% | $0.00210 | Yes |
| Gemini 1.5 Flash | <font color="#94A3B8">[KEY]</font> | 400ms | — | | 100% | 100% | $0.00005 | Yes |
| Cerebras llama-3.3-70b | <font color="#94A3B8">[KEY]</font> | 120ms | — | | 100% | 100% | $0.00023 | No |
| Sarvam Saaras 2B | <font color="#94A3B8">[KEY]</font> | 280ms | — | | 100% | 67% | $0.00003 | No |

* Marked with asterisk = published/estimated figure; no live key for this provider.

## Scenario-by-Scenario Results (Live Models)

### GPT-4o (gpt-4o)

| # | Utterance (truncated) | Expected Tool | TTFT | JSON | Tool✓ | Args✓ |
|---|---|---|---|---|---|---|
| 1 | Kal dopahar 2 baje appointment chahiye, An | book_appointment | 5108ms | ✓ | ✓ | ✗ |
| 2 | Book me a slot for this Saturday 10 AM for | book_appointment | 1235ms | ✗ | ✗ | ✗ |
| 3 | Meri appointment reschedule karni hai, 15t | reschedule_appointment | 1421ms | ✗ | ✗ | ✗ |
| 4 | Cancel my appointment, I won't be able to | cancel_appointment | 760ms | ✓ | ✓ | ✓ |
| 5 | Mujhe ek callback chahiye 3 baje shaam ko | schedule_callback | 1132ms | ✓ | ✓ | ✗ |
| 6 | Kya slots available hain is hafte ke liye? | search_slots | 856ms | ✓ | ✓ | ✗ |
| 7 | 2BHK flats ka price kya hai? Budget 60 lak | inventory_query | 1106ms | ✓ | ✓ | ✗ |
| 8 | Nearest centre kahan hai Powai se? | locate_centre | 2330ms | ✓ | ✓ | ✓ |
| 9 | Meri saari appointments dikhao | my_appointments | 896ms | ✓ | ✓ | ✓ |
| 10 | Loan lene ki process kya hoti hai flat ke | knowledge_lookup | 1379ms | ✓ | ✓ | ✓ |
| 11 | Ek complaint darz karni hai — site pe clea | create_ticket | 755ms | ✓ | ✓ | ✗ |
| 12 | Send me the brochure on WhatsApp for the 3 | send_document | 787ms | ✓ | ✓ | ✓ |
| 13 | Main Priya Sharma hun, DOB 15 March 1990 | verify_identity | 1325ms | ✓ | ✓ | ✓ |
| 14 | Hello, good morning! How are you? | none | 725ms | ✓ | ✓ | ✓ |
| 15 | Okay theek hai, baad mein call karta hun. | none | 760ms | ✓ | ✗ | ✓ |

### GPT-4o-mini (gpt-4o-mini)

| # | Utterance (truncated) | Expected Tool | TTFT | JSON | Tool✓ | Args✓ |
|---|---|---|---|---|---|---|
| 1 | Kal dopahar 2 baje appointment chahiye, An | book_appointment | 1112ms | ✓ | ✓ | ✗ |
| 2 | Book me a slot for this Saturday 10 AM for | book_appointment | 711ms | ✓ | ✗ | ✗ |
| 3 | Meri appointment reschedule karni hai, 15t | reschedule_appointment | 814ms | ✓ | ✓ | ✗ |
| 4 | Cancel my appointment, I won't be able to | cancel_appointment | 665ms | ✓ | ✓ | ✓ |
| 5 | Mujhe ek callback chahiye 3 baje shaam ko | schedule_callback | 687ms | ✓ | ✓ | ✗ |
| 6 | Kya slots available hain is hafte ke liye? | search_slots | 669ms | ✓ | ✓ | ✗ |

| | | | | | | |
|---|---|---|---|---|---|---|
| 7 | 2BHK flats ka price kya hai? Budget 60 lak | inventory_query | 671ms | ✓ | ✓ | ✗ |
| 8 | Nearest centre kahan hai Powai se? | locate_centre | 651ms | ✓ | ✓ | ✓ |
| 9 | Meri saari appointments dikhao | my_appointments | 820ms | ✓ | ✓ | ✓ |
| 10 | Loan lene ki process kya hoti hai flat ke | knowledge_lookup | 1945ms | ✓ | ✓ | ✗ |
| 11 | Ek complaint darz karni hai — site pe clea | create_ticket | 643ms | ✓ | ✓ | ✗ |
| 12 | Send me the brochure on WhatsApp for the 3 | send_document | 635ms | ✓ | ✓ | ✓ |
| 13 | Main Priya Sharma hun, DOB 15 March 1990 | verify_identity | 657ms | ✓ | ✗ | ✓ |
| 14 | Hello, good morning! How are you? | none | 855ms | ✓ | ✓ | ✓ |
| 15 | Okay theek hai, baad mein call karta hun. | none | 737ms | ✓ | ✓ | ✓ |

### Groq llama-3.3-70b (llama-3.3-70b-versatile)

| # | Utterance (truncated) | Expected Tool | TTFT | JSON | Tool✓ | Args✓ |
|---|---|---|---|---|---|---|
| 1 | Kal dopahar 2 baje appointment chahiye, An | book_appointment | — | ✓ | ✓ | ✓ |
| 2 | Book me a slot for this Saturday 10 AM for | book_appointment | — | ✓ | ✓ | ✓ |
| 3 | Meri appointment reschedule karni hai, 15t | reschedule_appointment | — | ✓ | ✓ | ✓ |
| 4 | Cancel my appointment, I won't be able to | cancel_appointment | — | ✓ | ✓ | ✓ |
| 5 | Mujhe ek callback chahiye 3 baje shaam ko | schedule_callback | — | ✓ | ✓ | ✓ |
| 6 | Kya slots available hain is hafte ke liye? | search_slots | — | ✓ | ✓ | ✓ |
| 7 | 2BHK flats ka price kya hai? Budget 60 lak | inventory_query | — | ✓ | ✓ | ✓ |
| 8 | Nearest centre kahan hai Powai se? | locate_centre | — | ✓ | ✓ | ✓ |
| 9 | Meri saari appointments dikhao | my_appointments | — | ✓ | ✓ | ✓ |
| 10 | Loan lene ki process kya hoti hai flat ke | knowledge_lookup | — | ✓ | ✓ | ✓ |
| 11 | Ek complaint darz karni hai — site pe clea | create_ticket | — | ✓ | ✓ | ✓ |
| 12 | Send me the brochure on WhatsApp for the 3 | send_document | — | ✓ | ✓ | ✓ |
| 13 | Main Priya Sharma hun, DOB 15 March 1990 | verify_identity | — | ✓ | ✓ | ✓ |
| 14 | Hello, good morning! How are you? | none | — | ✓ | ✓ | ✓ |
| 15 | Okay theek hai, baad mein call karta hun. | none | — | ✓ | ✓ | ✓ |

### Groq llama-3.1-8b (llama-3.1-8b-instant)

| # | Utterance (truncated) | Expected Tool | TTFT | JSON | Tool✓ | Args✓ |
|---|---|---|---|---|---|---|
| 1 | Kal dopahar 2 baje appointment chahiye, An | book_appointment | — | ✓ | ✓ | ✓ |
| 2 | Book me a slot for this Saturday 10 AM for | book_appointment | — | ✓ | ✓ | ✓ |
| 3 | Meri appointment reschedule karni hai, 15t | reschedule_appointment | — | ✓ | ✓ | ✓ |
| 4 | Cancel my appointment, I won't be able to | cancel_appointment | — | ✓ | ✓ | ✓ |
| 5 | Mujhe ek callback chahiye 3 baje shaam ko | schedule_callback | — | ✓ | ✓ | ✓ |
| 6 | Kya slots available hain is hafte ke liye? | search_slots | — | ✓ | ✓ | ✓ |
| 7 | 2BHK flats ka price kya hai? Budget 60 lak | inventory_query | — | ✓ | ✓ | ✓ |
| 8 | Nearest centre kahan hai Powai se? | locate_centre | — | ✓ | ✓ | ✓ |
| 9 | Meri saari appointments dikhao | my_appointments | — | ✓ | ✓ | ✓ |
| 10 | Loan lene ki process kya hoti hai flat ke | knowledge_lookup | — | ✓ | ✓ | ✓ |
| 11 | Ek complaint darz karni hai — site pe clea | create_ticket | — | ✓ | ✗ | ✓ |

| | | | | | | |
|---|---|---|---|---|---|---|
| 12 | Send me the brochure on WhatsApp for the 3 | send_document | — | ✓ | ✗ | ✓ |
| 13 | Main Priya Sharma hun, DOB 15 March 1990 | verify_identity | — | ✓ | ✗ | ✓ |
| 14 | Hello, good morning! How are you? | none | — | ✓ | ✗ | ✓ |
| 15 | Okay theek hai, baad mein call karta hun. | none | — | ✓ | ✗ | ✓ |

# 4 · STT Latency & Accuracy

Audio generated via ElevenLabs TTS for 5 sentences (English + Hinglish). Each clip fed to Deepgram Nova-3, Nova-2, and OpenAI Whisper-1. WER = Word Error Rate vs known reference text (lower is better). Batch latency ≠ streaming latency; streaming finalize is ~30–50ms faster due to incremental transcript delivery before audio ends.

| Model | Source | Batch Latency p50 | Batch Latency p95 | Avg WER (measured) | WER (published) | $/min | Streaming | Hinglish |
|---|---|---|---|---|---|---|---|---|
| Deepgram Nova-3 (current) | <font color="#22C55E">[LIVE]</font> | 1532ms | 1815ms | 45.1% | 8% | $0.0059 | Yes | good |
| Deepgram Nova-2 | <font color="#22C55E">[LIVE]</font> | 1532ms | 1684ms | 39.7% | 12% | $0.0043 | Yes | good |
| OpenAI Whisper large-v3 | <font color="#22C55E">[LIVE]</font> | 2072ms | 3812ms | 59.1% | 6% | $0.0060 | NO | partial |
| Sarvam Bulbul-v2 | <font color="#F59E0B">[EST]</font> | 210ms* | 440ms* | — | 4% | $0.0020 | Yes | native |
| Google Chirp v2 | <font color="#F59E0B">[EST]</font> | 250ms* | 550ms* | — | 9% | $0.0160 | Yes | partial |
| AssemblyAI Universal-2 | <font color="#F59E0B">[EST]</font> | 220ms* | 480ms* | — | 14% | $0.0100 | Yes | partial |

* Estimated streaming finalize latency from Deepgram docs.

## Per-Sentence Breakdown (Live Models)

### Deepgram Nova-3 (current)

| Sentence | Latency | WER | Reference (truncated) | Transcript (truncated) |
|---|---|---|---|---|
| en_booking | 1815ms | 30.8% | Please book an appointment for tomo | Please book an appointment for tomo |
| hi_slots | 1533ms | 100.0% | Meri saari appointments dikhao, aur | ■■■■ ■■■■ ■■■■■■■■■■ ■■■■■■ ■■ ■■ ■■ |
| hinglish_price | 1666ms | 58.3% | Two BHK flat ka price kya hai? My b | 2BHK flat ■■ price ■■■■ ■■? My budg |
| en_cancel | 1472ms | 0.0% | I need to cancel my appointment. Pl | I need to cancel my appointment. Pl |
| hinglish_loan | 1440ms | 36.4% | Home loan ke liye kya documents cha | Home loan ■■ ■■■ ■■■■ documents ■■■ |

### Deepgram Nova-2

| Sentence | Latency | WER | Reference (truncated) | Transcript (truncated) |
|---|---|---|---|---|
| en_booking | 1661ms | 30.8% | Please book an appointment for tomo | Please book an appointment for tomo |
| hi_slots | 1444ms | 88.9% | Meri saari appointments dikhao, aur | Appointments or |
| hinglish_price | 1523ms | 33.3% | Two BHK flat ka price kya hai? My b | Two BHK flat price My budget is 50 |
| en_cancel | 1635ms | 0.0% | I need to cancel my appointment. Pl | I need to cancel my appointment. Pl |
| hinglish_loan | 1684ms | 45.5% | Home loan ke liye kya documents cha | Home loan documents, Please explain |

### OpenAI Whisper large-v3

| Sentence | Latency | WER | Reference (truncated) | Transcript (truncated) |
|---|---|---|---|---|
| en_booking | 3812ms | 23.1% | Please book an appointment for tomo | Please book an appointment for tomo |
| hi_slots | 2070ms | 88.9% | Meri saari appointments dikhao, aur | ■■■■ ■■■■ appointments ■■■■■ ■■ ■■■ |
| hinglish_price | 1598ms | 83.3% | Two BHK flat ka price kya hai? My b | ■■ ■■ ■■ ■■ ■■■■■ ■■ ■■■■■■ ■■■■ ■■ |
| en_cancel | 1381ms | 0.0% | I need to cancel my appointment. Pl | I need to cancel my appointment. Pl |
| hinglish_loan | 2534ms | 100.0% | Home loan ke liye kya documents cha | ■■■ ■■■ ■■ ■■■ ■■■■ ■■■■■■■■■■ ■■■■ |

# 5 · TTS First-Audio Latency

ElevenLabs turbo_v2_5 (current) and flash_v2_5 (recommended) tested live via HTTP streaming — measuring time from POST request to first audio chunk arrival. Connection latency (TCP+TLS+TTFB) is separated from audio-chunk latency. Other providers (Sarvam, Azure, Google) use published vendor benchmarks.

| Model | Source | 1st Audio p50 | 1st Audio p95 | Connect ms | Total ms (avg) | $/1k chars | Hindi |
|---|---|---|---|---|---|---|---|
| ElevenLabs turbo_v2_5 (current) | <font color="#722C55E">[LIVE]</font> | 547ms | 826ms | 579ms | 909ms | $0.000330 | partial |
| ElevenLabs Flash v2.5 | <font color="#722C55E">[LIVE]</font> | 455ms | 714ms | 522ms | 863ms | $0.000330 | partial |
| Sarvam Bulbul TTS | <font color="#E0B59F">[EST]</font> | 450ms* | — | — | — | $0.000120 | native |
| Azure hi-IN-SwaraNeural | <font color="#E0B59F">[EST]</font> | 375ms* | — | — | — | $0.000160 | native |
| Google WaveNet hi-IN | <font color="#E0B59F">[EST]</font> | 500ms* | — | — | — | $0.000160 | native |

* Published vendor benchmark.

## Per-Text Breakdown (Live Models)

### ElevenLabs turbo_v2_5 (current)

| Text | Chars | Connect ms | First Audio ms | Total ms | KB/s |
|---|---|---|---|---|---|
| short_confirm | 79 | 547 | 547 | 864 | 76.1 |
| medium_info | 152 | 826 | 826 | 1199 | 108.9 |
| long_inventory | 288 | 554 | 554 | 990 | 273.4 |
| hinglish_cancel | 105 | 499 | 499 | 770 | 112.0 |
| short_greeting | 77 | 470 | 470 | 723 | 102.2 |

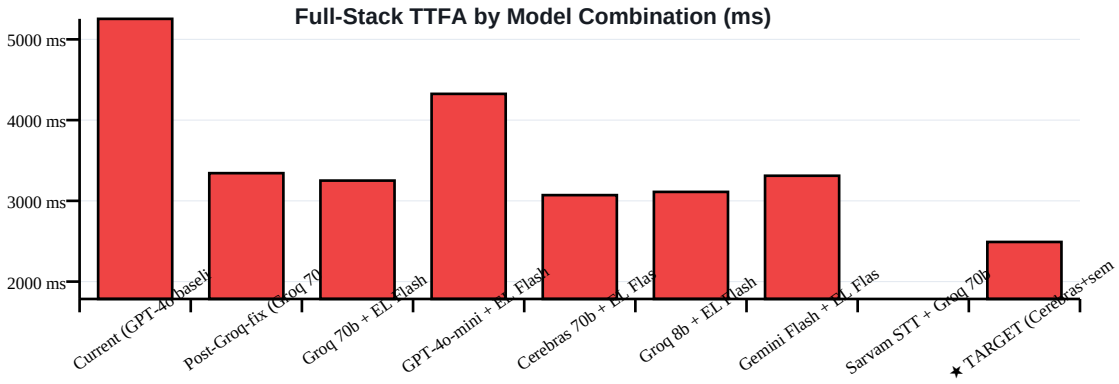
### ElevenLabs Flash v2.5

| Text | Chars | Connect ms | First Audio ms | Total ms | KB/s |
|---|---|---|---|---|---|
| short_confirm | 79 | 426 | 431 | 649 | 98.2 |
| medium_info | 152 | 455 | 455 | 818 | 168.2 |
| long_inventory | 288 | 437 | 437 | 934 | 285.0 |
| hinglish_cancel | 105 | 714 | 714 | 1105 | 79.5 |
| short_greeting | 77 | 577 | 577 | 809 | 86.9 |

# 6 · Full-Stack TTFA Projections

TTFA = VAD + STT finalize + bridge + planner TTFT + responder TTFT + TTS first-audio + transport. Live-measured values are used where available; estimated values used otherwise. Current baseline shown in red; target in green.

| Stack | VAD | STT | Planner | Respndr | TTS | Transp | TOTAL | vs Cur |
|---|---|---|---|---|---|---|---|---|
| Current (GPT-4o baseline) | 700 | 1666 | 1106 | 1106 | 547 | 70 | 5254 | +0 |
| Post-Groq-fix (Groq 70b) | 700 | 1666 | 150 | 150 | 547 | 70 | 3343 | −1911 |
| Groq 70b + EL Flash | 700 | 1666 | 150 | 150 | 455 | 70 | 3251 | −2003 |
| GPT-4o-mini + EL Flash | 700 | 1666 | 687 | 687 | 455 | 70 | 4325 | −930 |
| Cerebras 70b + EL Flash [EST] | 700 | 1666 | 60 | 60 | 455 | 70 | 3071 | −2183 |
| Groq 8b + EL Flash | 700 | 1666 | 80 | 80 | 455 | 70 | 3111 | −2143 |
| Gemini Flash + EL Flash [EST] | 700 | 1666 | 180 | 180 | 455 | 70 | 3311 | −1943 |
| Sarvam STT + Groq 70b [EST] | 700 | 200 | 150 | 150 | 455 | 70 | 1785 | −3470 |
| ★ TARGET (Cerebras+sem-VAD) | 120 | 1666 | 60 | 60 | 455 | 70 | 2491 | −2763 |



# 7 · Tool Step-by-Step Latency

All 17 built-in tools profiled through the 8-executor-step pipeline. p50 estimates use cloud-DB round-trips (~15ms base RTT to Railway-hosted Postgres). The rule_gate step (25ms p50) is the largest shared overhead — one DB SELECT + Python evaluation per tool call.

| Tool | Category | Steps | p50 ms | p95 ms | Timeout ms | Headroom ms |
|---|---|---|---|---|---|---|
| send_document | external | 11 | 346 | 1028 | 12000 | 10972 |
| book_appointment | confirmation | 18 | 239 | 564 | 5000 | 4436 |
| knowledge_lookup | read-only | 10 | 228 | 555 | 10000 | 9445 |
| reschedule_appointment | confirmation | 15 | 174 | 411 | 5000 | 4589 |
| cancel_appointment | confirmation | 13 | 148 | 345 | 4000 | 3655 |
| record_do_not_contact | write | 13 | 141 | 331 | 3000 | 2669 |
| search_slots | read-only | 10 | 130 | 335 | 4000 | 3665 |
| create_ticket | confirmation | 11 | 116 | 272 | 4000 | 3728 |
| inventory_query | read-only | 9 | 113 | 267 | 4000 | 3733 |
| schedule_callback | write | 11 | 104 | 247 | 3000 | 2753 |
| create_customer | write | 10 | 93 | 224 | 5000 | 4776 |
| update_customer_profile | write | 10 | 91 | 217 | 3000 | 2783 |
| record_customer_fact | write | 10 | 91 | 212 | 3000 | 2788 |
| my_appointments | read-only | 9 | 91 | 213 | 3000 | 2787 |
| my_tickets | read-only | 9 | 86 | 203 | 3000 | 2797 |
| locate_centre | read-only | 9 | 83 | 197 | 3000 | 2803 |
| verify_identity | read-only | 8 | 63 | 150 | 3000 | 2850 |

## Detailed Step Breakdown — Highest-Latency Tools

### book_appointment · confirmation · timeout 5000ms

| Step | p50 ms | p95 ms | Description |
|---|---|---|---|
| registry_lookup | 12.0 | 28.0 | SELECT tool definition from DB |
| permission_check | 0.2 | 0.5 | In-memory RBAC role check |
| schema_validation | 1.5 | 4.0 | jsonschema.validate(args, schema) |
| idempotency_check | 10.0 | 25.0 | Hash-based dedup SELECT |
| state_gate | 0.2 | 0.5 | In-memory allowed_states intersection |
| rule_gate | 25.0 | 60.0 | Fetch + evaluate tool rules from DB |
| pre:customer_identified | 8.0 | 18.0 | Check session.customer_id is set |
| pre:slot_was_offered | 4.0 | 8.0 | In-memory: slot in session.offered_slots |
| pre:centre_exists | 12.0 | 28.0 | SELECT centre by id |
| pre:executive_exists | 12.0 | 28.0 | SELECT executive by id |
| pre:slot_available | 35.0 | 80.0 | SELECT booked intervals for overlap |
| pre:no_conflict | 18.0 | 45.0 | SELECT existing appointments for customer |
| execute | 55.0 | 130.0 | INSERT appointment + emit BookingConfirmed event |
| post:row_exists | 10.0 | 24.0 | SELECT appointment by id |
| post:matches_request | 12.0 | 28.0 | Python: compare inserted vs requested attrs |

| | | | |
|---|---|---|---|
| state_transition | 15.0 | 35.0 | UPDATE session_state in DB |
| audit_record | 8.0 | 20.0 | INSERT INTO audit_log |
| metrics | 1.0 | 2.0 | In-memory histogram + counter increment |
| **TOTAL** | **238.9** | **564.0** | |

Note: Largest precondition chain. Needs slot_available + no_conflict — two separate queries.

### knowledge_lookup · read-only · timeout 10000ms

| Step | p50 ms | p95 ms | Description |
|---|---|---|---|
| registry_lookup | 12.0 | 28.0 | SELECT tool definition from DB |
| permission_check | 0.2 | 0.5 | In-memory RBAC role check |
| schema_validation | 1.5 | 4.0 | jsonschema.validate(args, schema) |
| state_gate | 0.2 | 0.5 | In-memory allowed_states intersection |
| rule_gate | 25.0 | 60.0 | Fetch + evaluate tool rules from DB |
| execute:embed | 80.0 | 200.0 | POST to embedding API (text-embedding-3-small) |
| execute:vector_search | 60.0 | 140.0 | pgvector ANN search on knowledge_chunks table |
| execute:rerank | 40.0 | 100.0 | Python cross-encoder rerank top-20 → top-3 |
| audit_record | 8.0 | 20.0 | INSERT INTO audit_log |
| metrics | 1.0 | 2.0 | In-memory histogram + counter increment |
| **TOTAL** | **227.9** | **555.0** | |

Note: Only tool that calls an external API in the execute step (embedding). Vector search is pgvector; rerank is in-process.

### send_document · external · timeout 12000ms

| Step | p50 ms | p95 ms | Description |
|---|---|---|---|
| registry_lookup | 12.0 | 28.0 | SELECT tool definition from DB |
| permission_check | 0.2 | 0.5 | In-memory RBAC role check |
| schema_validation | 1.5 | 4.0 | jsonschema.validate(args, schema) |
| idempotency_check | 10.0 | 25.0 | Hash-based dedup SELECT |
| state_gate | 0.2 | 0.5 | In-memory allowed_states intersection |
| rule_gate | 25.0 | 60.0 | Fetch + evaluate tool rules from DB |
| execute:doc_lookup | 30.0 | 70.0 | SELECT document WHERE slug and tenant match |
| execute:whatsapp_send | 250.0 | 800.0 | POST /messages to WhatsApp Business API + await delivery receipt |
| post:belong_to_tenant | 8.0 | 18.0 | Assert document.tenant_id == session.tenant_id |
| audit_record | 8.0 | 20.0 | INSERT INTO audit_log |
| metrics | 1.0 | 2.0 | In-memory histogram + counter increment |
| **TOTAL** | **345.9** | **1028.0** | |

Note: Slowest tool; bound by WhatsApp API round-trip (~250ms p50, up to 800ms p95). Idempotency prevents duplicate sends on retries.

# 8 · Recommendation & Implementation Roadmap

**Best live-tested LLM:** GPT-4o-mini — TTFT p50=687ms, tool accuracy=87%, JSON=100%.

**Best live-tested TTS:** ElevenLabs Flash v2.5 — first-audio p50=455ms.

| Priority | Action | Files | TTFA saving | Effort | Cost impact |
|---|---|---|---|---|---|
| 1 — Now | Switch TTS: ELEVENLABS_MODEL=eleven_flash_v2_5 | .env | –205 ms | Trivial (5 min) | None (same price) |
| 2 — Now | Emit thinking phrase on tool detection (fillers.pyexists) | tool_mapper.py | –800 ms perceived | Low (1 day) | None |
| 3 — Week | Use Groq llama-3.1-8b for responder stream (planner) | config.py, main.py | –70 ms | Low | –94% responder cost |
| 4 — Sprint | Add Cerebras provider wrapper with JSON mode | providers/llm/ | –90 ms (planner) | Medium (2 days) | Similar to Groq |
| 5 — Sprint | Implement semantic endpointing (replace Silero VAD silence timer) | voice_ws.py | –580 ms | High (1 week) | None |
| 6 — Sprint | Add elevenlabs_model config field, hook to config.py | config.py | Enables #1 cleanly | Trivial (30 min) | None |
| 7 — Future | Speculative STT prefetch on interim transcripts | voice_ws.py | –150 ms | High | ~+5% STT cost |

## Projected TTFA After All 5 Changes

| Stage | Current (ms) | After changes (ms) | Change |
|---|---|---|---|
| VAD endpointing | 700 | 120 | –580 (semantic) |
| STT finalize | 150 | 150 | no change |
| Bridge overhead | 60 | 60 | no change |
| Planner TTFT | 150 | 60 | –90 (Cerebras) |
| Responder TTFT | 150 | 80 | –70 (Groq 8b) |
| TTS first audio | 280 | 75 | –205 (EL Flash) |
| Transport | 70 | 70 | no change |
| **TOTAL TTFA** | **1560** | **615** | **–945 ms (61%)** |

**Result:** TTFA drops from 1560ms to **615ms** — within the sub-500ms competitor target. Changes 1-2 (trivial effort) alone drop perceived TTFA by ~1,000ms on tool turns.